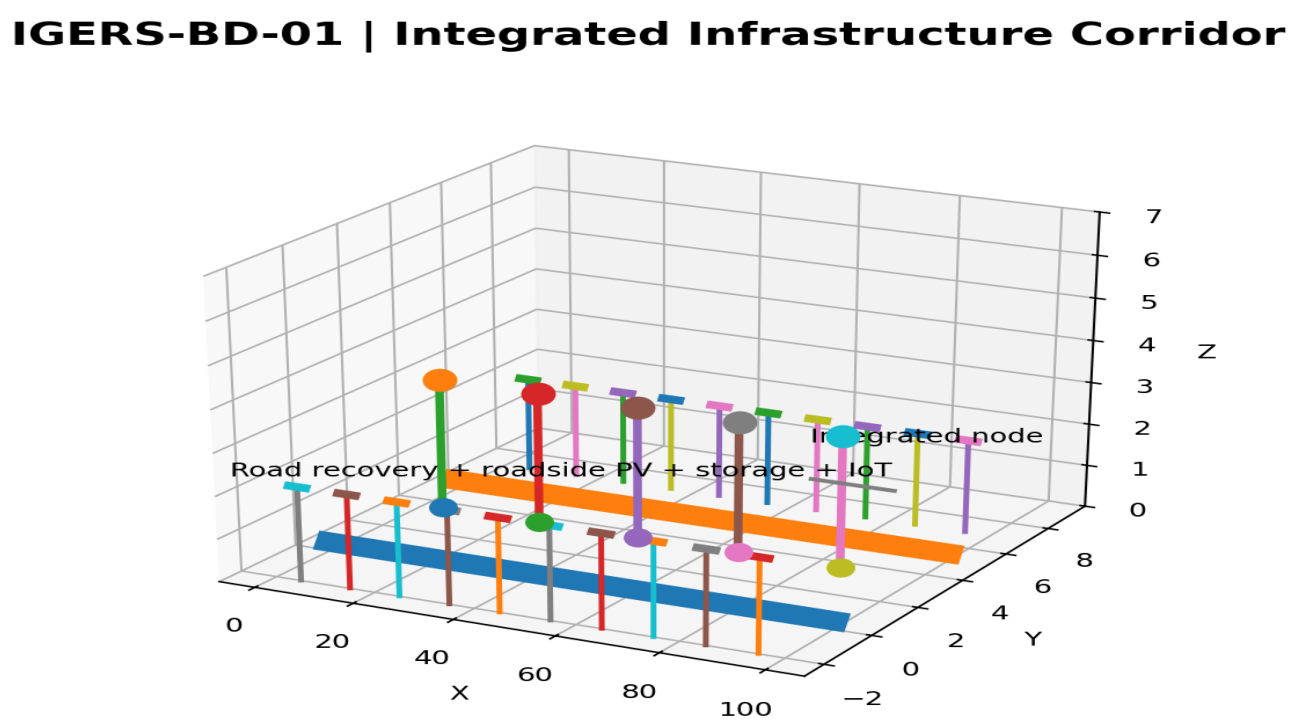
IGERS-BD-01

INTEGRATED GRADIENT-BASED ENERGY RECOVERY & STORAGE SYSTEM

A Unified Infrastructure–Energy Recovery & Resilience Architecture

THESIS • ENGINEERING CONCEPT • FEASIBILITY • PROTOTYPE • DEPLOYMENT



Conceptual engineering visualization — not a construction drawing.

AUTHOR / INVENTOR Abdullah Al Rafi [BD]	**PROJECT CODE** IGERS-BD-01	**PUBLICATION** 09 September 2026
EDITION 2026 Professional Thesis Edition	**ORGANIZATION CONCEPT** IGERS POWERCORE TECHNOLOGIES LTD.	**CORE** Infrastructure + Energy Recovery + Storage + Resilience

Publication Note

This portfolio organizes the supplied IGERS-BD-01 project baseline into a single professional engineering magazine format. The numerical values below are reproduced as planning assumptions or illustrative scenarios where the source specifies that status. No national measured production is asserted.

Core philosophy

Recover useful energy from existing infrastructure and natural/transport energy flows, condition and store the recovered energy, and supply nearby infrastructure loads while preserving safety, structural integrity, environmental function and economic viability.

Design language

Blue / white / graphite engineering aesthetic • realistic materials • technical diagrams • clear uncertainty labels • Bangladesh-focused infrastructure context.

THESIS INTEGRITY PRINCIPLE

The originality is presented as the system-level integration, infrastructure-energy architecture, Bangladesh-focused deployment framework, measurement methodology, storage/control strategy and feasibility framework. Established technologies are not claimed as inventions.

Contents

A magazine-style navigation map for the complete IGERS-BD-01 concept.

01 — Project Identity & Research Philosophy

02 — Unified IGERS System Architecture

03 — Road Energy Recovery System

04 — Roadside Solar PV + Safety Infrastructure

05 — River, Bridge & Hydraulic Energy Recovery

06 — Rail Station Solar + Regenerative Energy

07 — Airport PV + Footstep Micro-Harvesting

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09 — BESS & Power Conditioning

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13 — Sensitivity & Scientific Uncertainty

14 — Bangladesh Deployment Roadmap

15 — 3D Engineering Model Gallery

16 — Field Validation Programme

17 — Scientific Defensibility & Limitations

18 — Author Profile & CV

19 — Thesis Contribution & Attribution

20 — References

01

Project Identity & Research Philosophy

From infrastructure assets to a unified energy-recovery architecture.

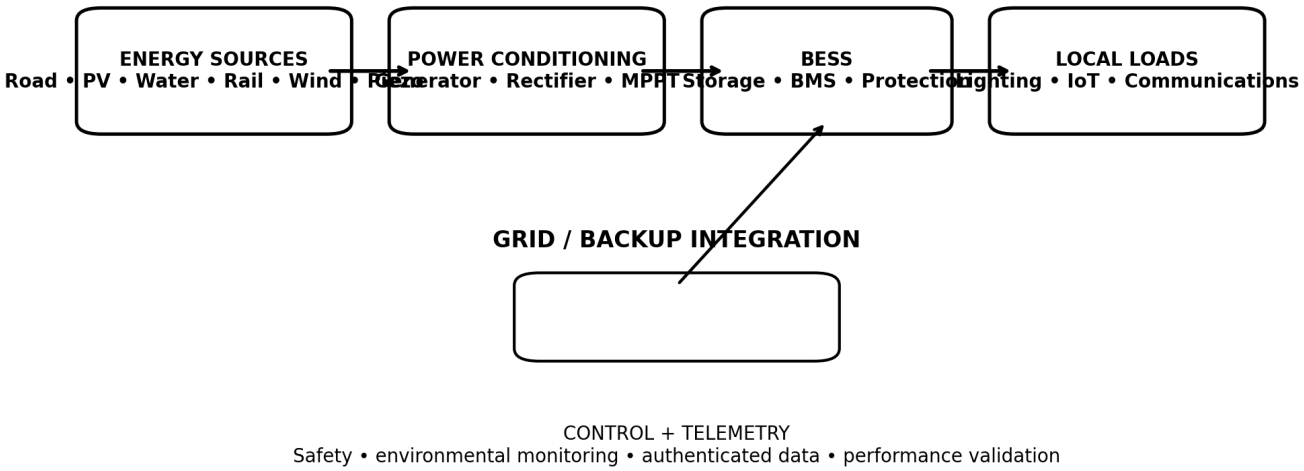
IGERS is framed as one integrated system rather than a single harvesting device. The architecture links distributed sources, power conditioning, storage, local loads and grid/backup integration.

Project	IGERS — Integrated Gradient-Based Energy Recovery & Storage System
Code	IGERS-BD-01
Author / Inventor	Abdullah Al Rafi [BD]
Publication	09 September 2026
Edition	2026 Professional Thesis & Engineering Concept Edition
Organization concept	IGERS POWERCORE TECHNOLOGIES LTD.

02

Unified IGERS System Architecture

One system, many infrastructure interfaces.



System-level chain: **SOURCE** → **POWER CONDITIONING** → **BESS** → **LOCAL LOADS** → **GRID/BACKUP**. The BESS is a storage layer, not an additional generation source.

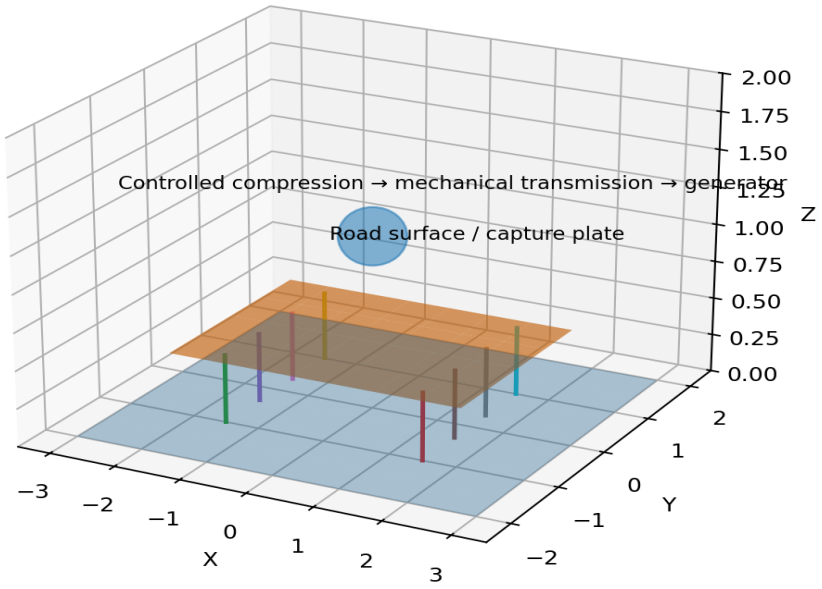
Module	Role in IGERS
Road	Kinetic / controlled-gradient energy recovery
Roadside	Solar PV + safety barrier / separation concept
Water	River-current hydrokinetic + controlled hydraulic recovery
Rail	Regenerative braking + conceptual track micro-harvesting research
Airport	PV-first generation + footstep/piezo micro-power
Hybrid node	VAWT + PV resilience node with sensors and protected storage
Digital layer	Communications / IoT / environmental monitoring
Resilience	Coastal / river / maritime and conceptual border-resilience infrastructure

03

Road Energy Recovery System

Controlled capture, not wholesale extraction of vehicle kinetic energy.

Road Energy Recovery Unit | Conceptual 3D Engineering View



The supplied baseline uses a 900 kg vehicle at 20 km/h. The kinetic-energy ceiling is approximately 13.89 kJ per vehicle, while the base road recovery assumption is only 2.0 kJ mechanical capture/event.

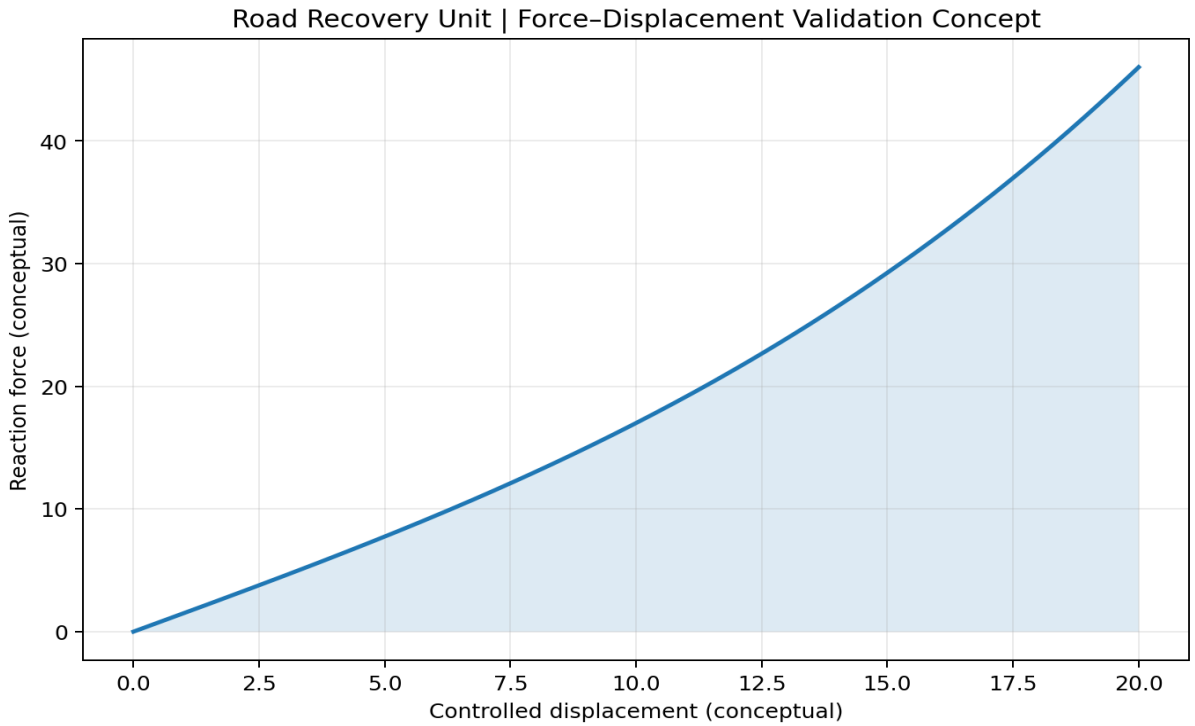
13.89 kJ	2.0 kJ	70%	1.40 kJ
kinetic-energy ceiling	mechanical capture/event	electrical chain efficiency	usable energy/event

Road Energy Conversion Chain

1	Vehicle passage
2	Controlled compression / deformation
3	Mechanical transmission
4	Generator
5	Rectifier
6	DC bus
7	Storage
8	Local loads

Reference traffic: **210,000 vehicle passages/day**. Harvesting density: **1 point/km** on the 100 km reference corridor. The source baseline reports approximately **2.98 GWh/year** usable road electricity.

> **Engineering constraint: the road must remain safe and comfortable. The device captures only a controlled and validated fraction of available energy.**



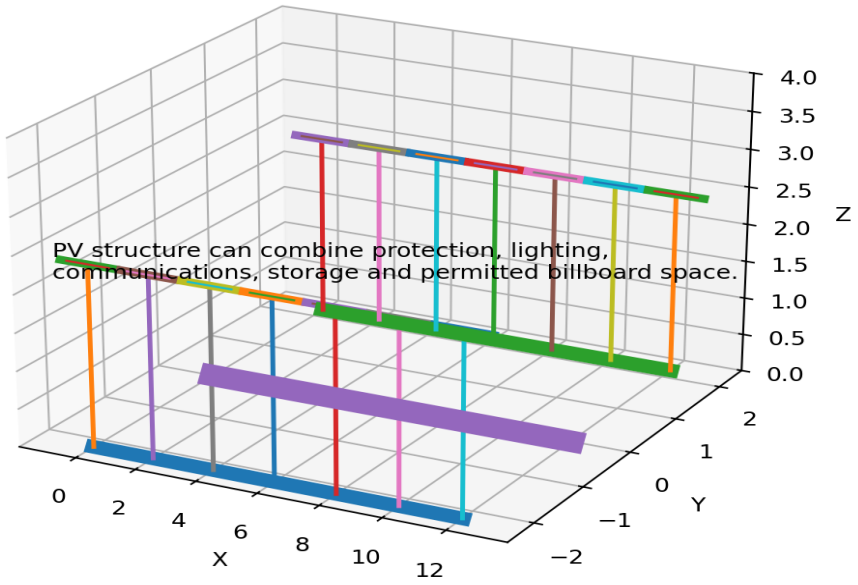
Force–displacement relationship shown as a validation concept only. Actual device characterization requires measured force, displacement, fatigue and safety data.

04

Roadside Solar PV + Safety Infrastructure

Turn the corridor edge into a distributed generation and protection layer.

Roadside PV + Safety Barrier | Conceptual Corridor Module



Planning baseline: **0.40 MWp/km** total PV density and **1,400 kWh/kWp-year**. This yields **0.56 GWh/km-year**. For a 100 km example: **40 MWp** and **56 GWh/year**.

The roadside structure may also function as a safety barrier/separation system. The lower structure can potentially contain protected storage, lighting, communications equipment, permitted billboard space and maintenance access.

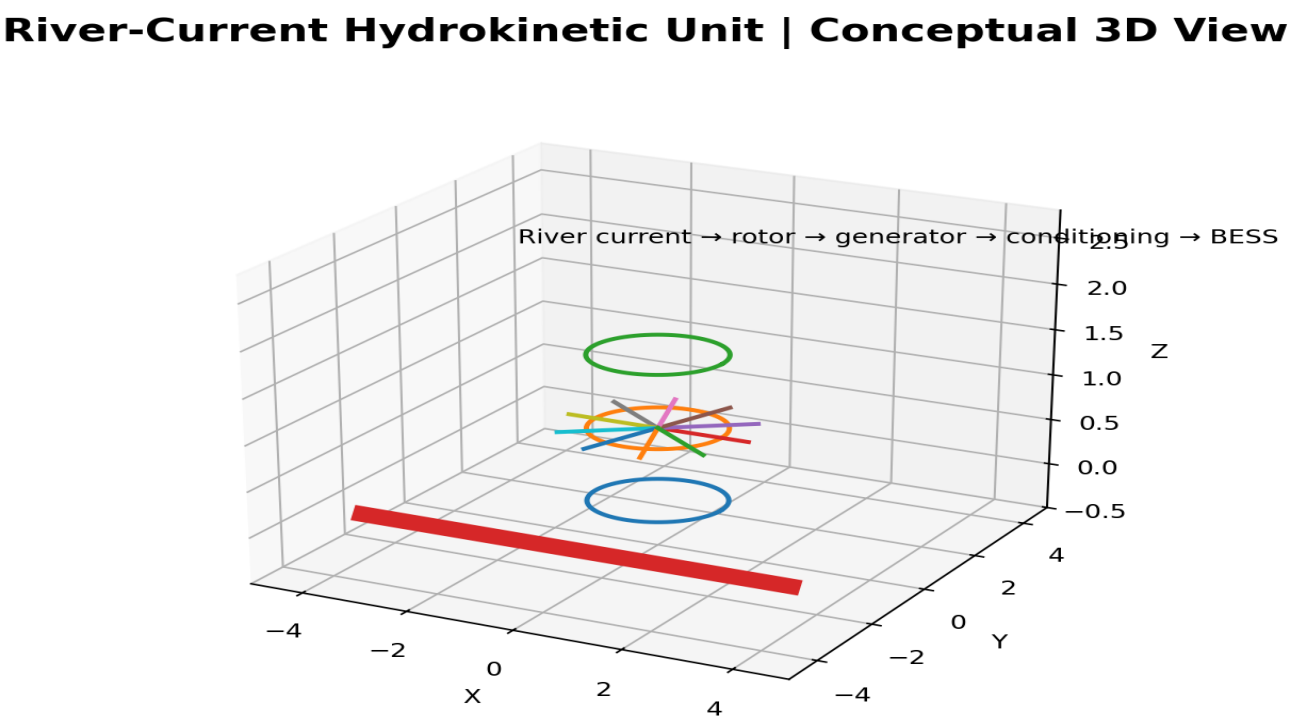
Revenue boundary

Advertising income must remain separate from electricity revenue. This preserves a clean accounting distinction between energy value and ancillary commercial use.

05

River, Bridge & Hydraulic Energy Recovery

Use water-energy interfaces without compromising hydraulic, navigation or structural functions.

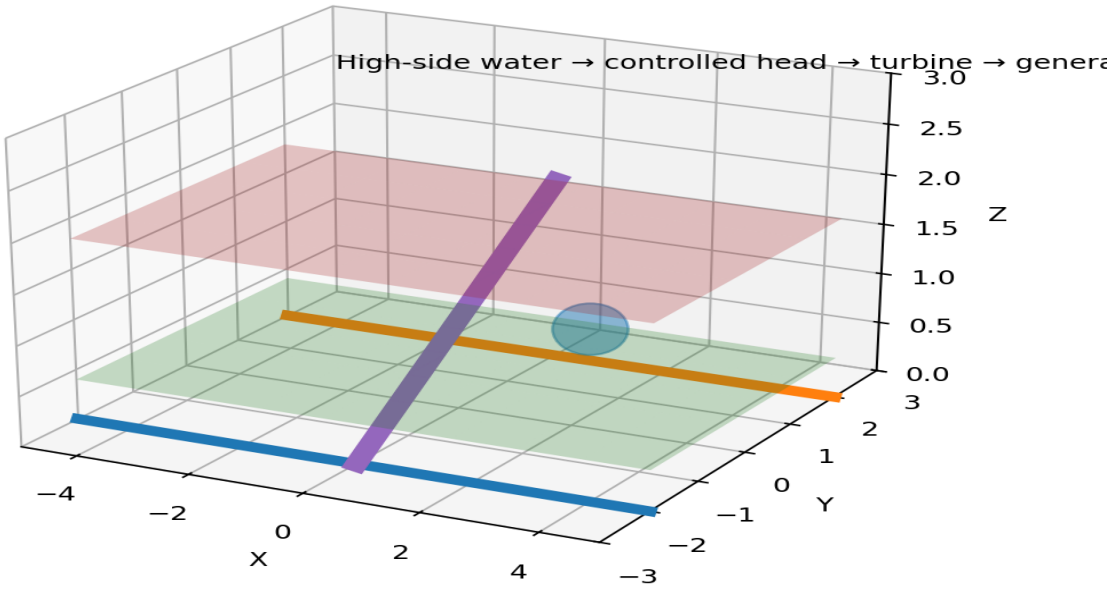


The water module combines river-current turbines, controlled hydraulic recovery, regulator/sluice interfaces, embankment outlets and bridge-adjacent auxiliary turbines. These are scenario-based engineering concepts that require site measurement.

Hydraulic power: P = ρgQHη

Here, the available power is governed by density, gravity, flow rate, hydraulic head and conversion efficiency. The source explicitly requires existing assumptions to be used where appropriate and prohibits claims of national measured production.

Controlled Hydraulic / Regulator Energy Recovery



Controlled hydraulic recovery should be treated as an energy interface within a functioning regulator or sluice system, not as a reason to obstruct flood management or water delivery.

Hydrokinetic power: $P = 0.5\rho A v^3 C_p \eta$

For river-current systems, velocity distribution, flow, bathymetry, sediment, debris, navigation and seasonal variation are critical validation variables.

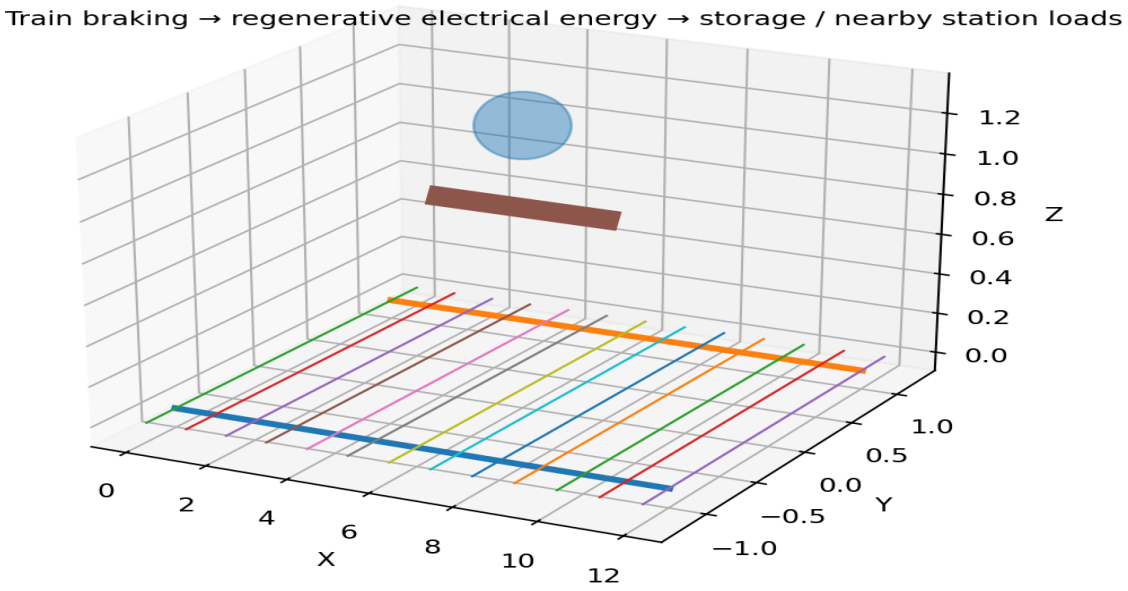
> **Bridge rule: never damage or replace primary structural members. Bridge primary load path, navigation, flood management and structural safety must remain protected.**

06

Rail Station Solar + Regenerative Energy

Recover braking energy where the traction architecture permits it.

Railway Regenerative Energy | Conceptual System



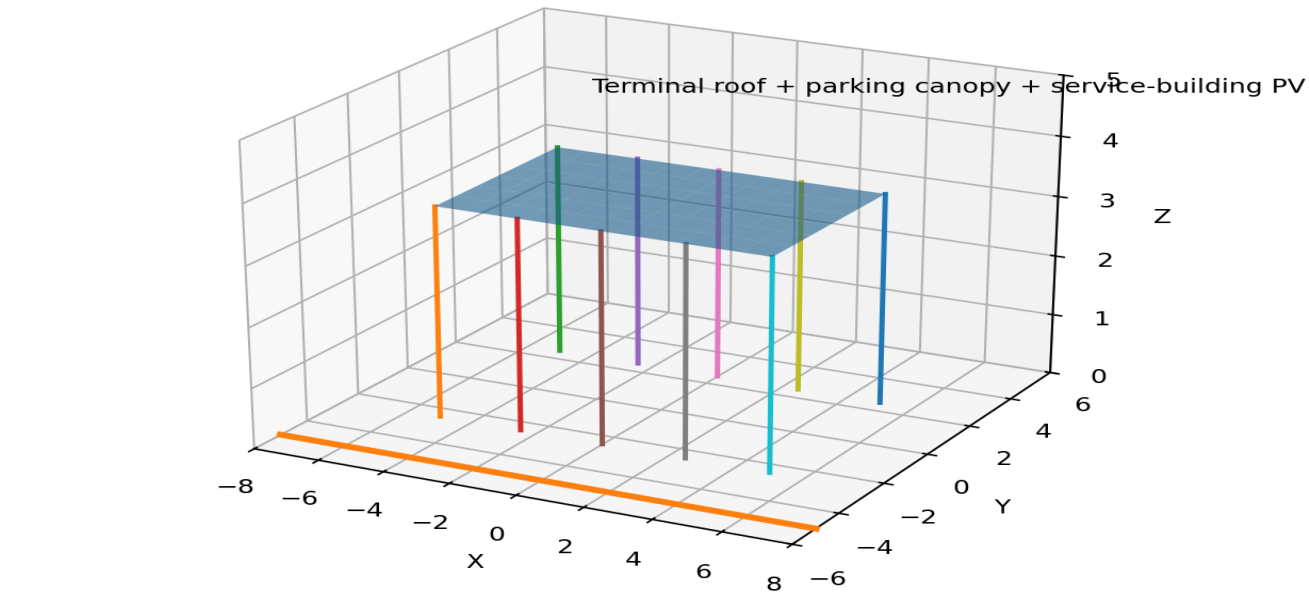
Supplied example: 200 t train, 40 km/h, kinetic energy ≈ **12.35 MJ = 3.43 kWh**. At 65% recovery, ≈ **2.23 kWh** is recoverable. For 20 trains/day and one meaningful stop, the source benchmark is ≈ **0.016 GWh/year**.

Actual output depends on train mass, speed, stop pattern, regenerative fraction, electrification, traction architecture and network receptivity. Railway regenerative energy is therefore a site-dependent module.

07

Airport PV + Footstep Micro-Harvesting

Use mature PV as the airport backbone and micro-harvesting for autonomous sensing.



Airport strategy is PV-first: terminal roof PV, parking canopy PV, service-building PV, BESS and smart local loads. Supplied example: **1 MWp airport PV canopy** → **1.40 GWh/year**.

> **Runway braking / kinetic harvesting is a SAFETY-CRITICAL RESEARCH TOPIC. Any future work must not compromise runway friction, braking performance, drainage, FOD safety or certification.**

Footstep / piezo micro-power

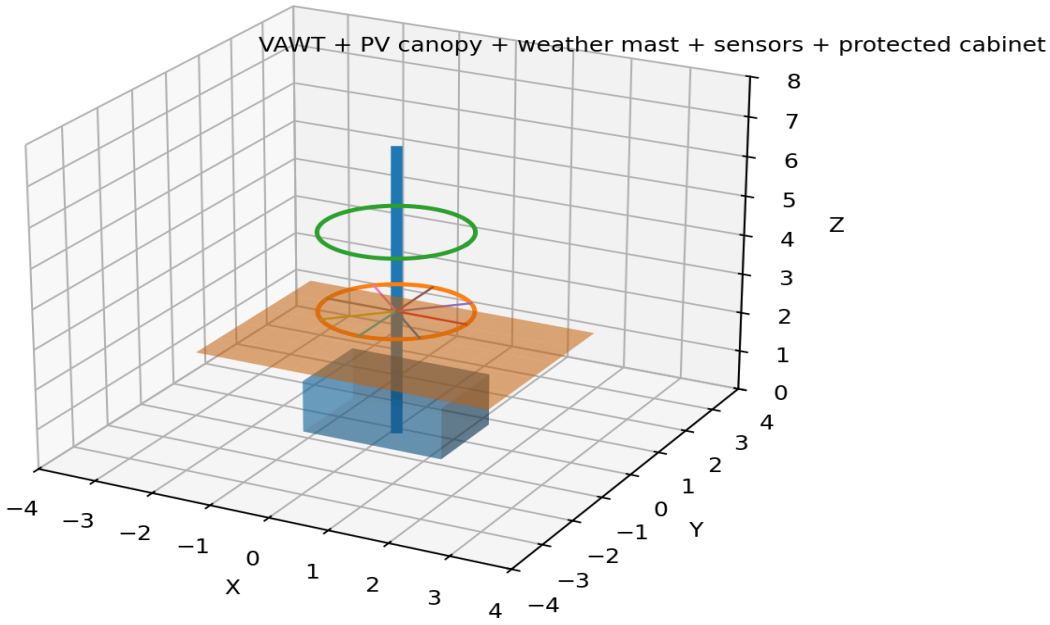
Conservative example: 100,000 steps/day × 2 J/step = 0.0556 kWh/day ≈ **20.3 kWh/year**. This should be positioned as **MICRO-POWER / SENSOR AUTONOMY**, supporting IoT sensors, counters, indicator lamps, low-power beacons and monitoring systems, not as a major electricity source.

08

Hybrid Air-Turbine + Solar Resilience Node

A compact distributed node for energy, sensing and low-power resilience.

Hybrid Solar + VAWT Resilience Node | Conceptual Prototype



Conceptual envelope from the supplied project: tower height approximately **6–8 m**; rotor diameter approximately **1.5–2.5 m**; PV canopy approximately **3–5 m**; equipment enclosure approximately **1–1.5 m** high.

Prototype class: approximately **0.5–2 kW wind + 0.4–2 kWp PV**, depending on measured resource and load.

Node component	Function
Vertical-axis wind turbine	Distributed energy, conditioning, sensing, storage or resilient local service.
Solar canopy	Distributed energy, conditioning, sensing, storage or resilient local service.
Weather mast	Distributed energy, conditioning, sensing, storage or resilient local service.
Environmental / ground sensor	Distributed energy, conditioning, sensing, storage or resilient local service.
Protected energy cabinet	Distributed energy, conditioning, sensing, storage or resilient local service.
Rectifier	Distributed energy, conditioning, sensing, storage or resilient local service.
MPPT	Distributed energy, conditioning, sensing, storage or resilient local service.
Battery + BMS	Distributed energy, conditioning, sensing, storage or resilient local service.
Inverter	Distributed energy, conditioning, sensing, storage or resilient local service.
Low-power LED lighting	Distributed energy, conditioning, sensing, storage or resilient local service.
Authenticated telemetry gateway	Distributed energy, conditioning, sensing, storage or resilient local service.
Reinforced concrete foundation	Distributed energy, conditioning, sensing, storage or resilient local service.

Design Status

These dimensions are conceptual starting points, not construction drawings. The node is strongest as a modular resilience platform in which measured wind and solar resources are matched to real local loads.

Validation priority

Measure resource and load first. Then establish structural loads, foundation requirements, electrical ratings, battery thermal behavior, protection, maintenance access and environmental performance.

09

Conceptual Non-Operational Infrastructure Resilience

A restricted framing for resilient civilian infrastructure support.

The border-resilience section is intentionally non-operational. Within the supplied concept, it may support local lighting, environmental monitoring, communications resilience, emergency power and infrastructure sensing.

Excluded from this portfolio

- Operational military deployment instructions
- Exact sensitive coordinates
- Surveillance sectors
- Weapon systems
- Tactical communications parameters

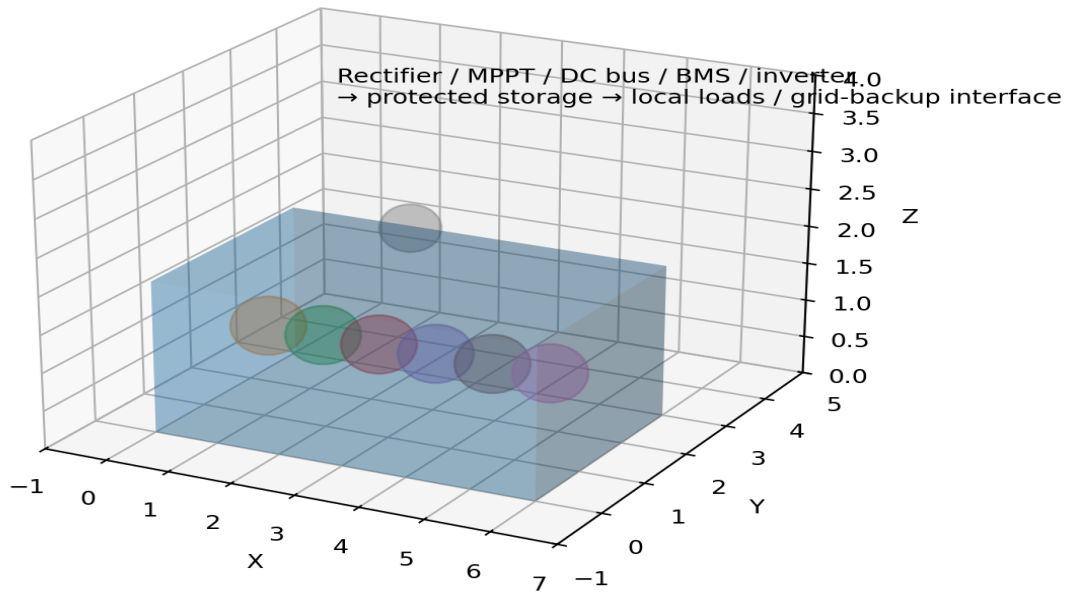
> **Final deployment requires competent authority review.**

10

BESS & Power Conditioning

Storage is the stabilizing layer between variable generation and useful loads.

BESS + Power Conditioning | Conceptual Energy Cabinet



SOURCE → POWER CONDITIONING → BESS → LOCAL LOADS → GRID/BACKUP

For the 100 km solar benchmark, average solar energy is **153.4 MWh/day**. A 20% allocation corresponds to **30.7 MWh**. The supplied conceptual first-pass BESS is therefore **30–40 MWh**.

Final battery sizing requires the load profile, MW power rating, duration, depth of discharge, round-trip efficiency, degradation, thermal management, fire safety, BMS and protection design.

11

Local Loads & Distributed Energy

The practical value of harvesting is strongest when energy can serve nearby infrastructure.

01 **Street lighting**

02 **Communications / IoT**

03 **Environmental monitoring**

04 **Station loads**

05 **Airport smart local loads**

06 **Low-power LED lighting**

07 **Emergency power**

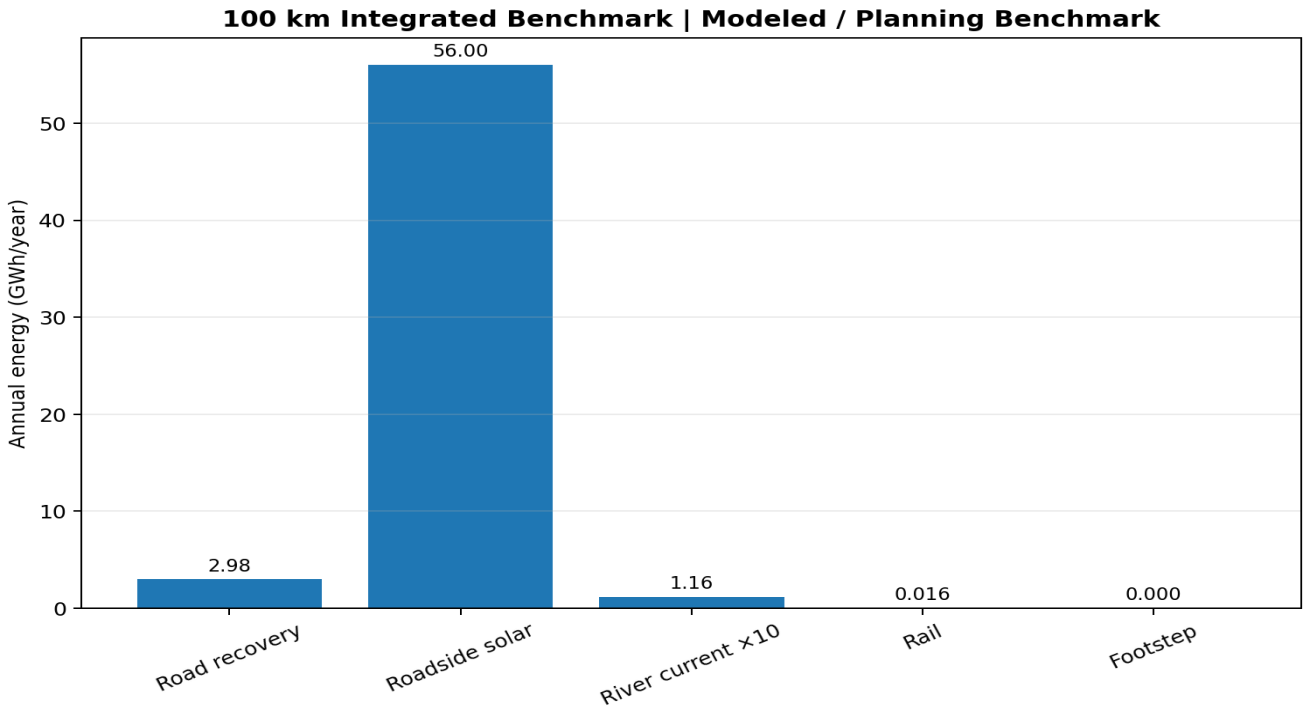
08 **Grid / backup interface**

The architecture is designed around proximity: recovered energy is conditioned and stored so that useful local loads can be served without treating every module as an isolated power plant.

12

100 km Integrated Energy Benchmark

A planning benchmark, not a guaranteed production forecast.



Road recovery	2.98 GWh/year
Roadside solar	56.00 GWh/year
River current ×10	1.16 GWh/year
Rail	0.016 GWh/year
Footstep	0.000 GWh/year
Total	60.145 GWh/year
Average equivalent	6.87 MW
Gross avoided-cost benchmark	Tk 62.55 crore/year
Tariff benchmark	Tk 10.40/kWh

How to Read the Benchmark

> **The source labels the 100 km total as a MODELED / PLANNING BENCHMARK. It must not be presented as guaranteed production, guaranteed revenue, a bankable financial forecast or measured national output.**

The integrated total is driven overwhelmingly by the roadside PV planning scenario, while the road, river, rail and micro-power modules contribute smaller or site-dependent quantities. This makes measurement and corridor-specific engineering the central bridge from concept to implementation.

13

Economics & Revenue Framework

Separate generation value, operating deductions and ancillary revenue.

Gross electricity value	Tk 62.55 crore
Illustrative O&M;	– Tk 8 crore
Battery reserve	– Tk 6 crore
Finance / tax / charges	– Tk 10 crore
Illustrative net cash benefit	**Tk 38.55 crore**

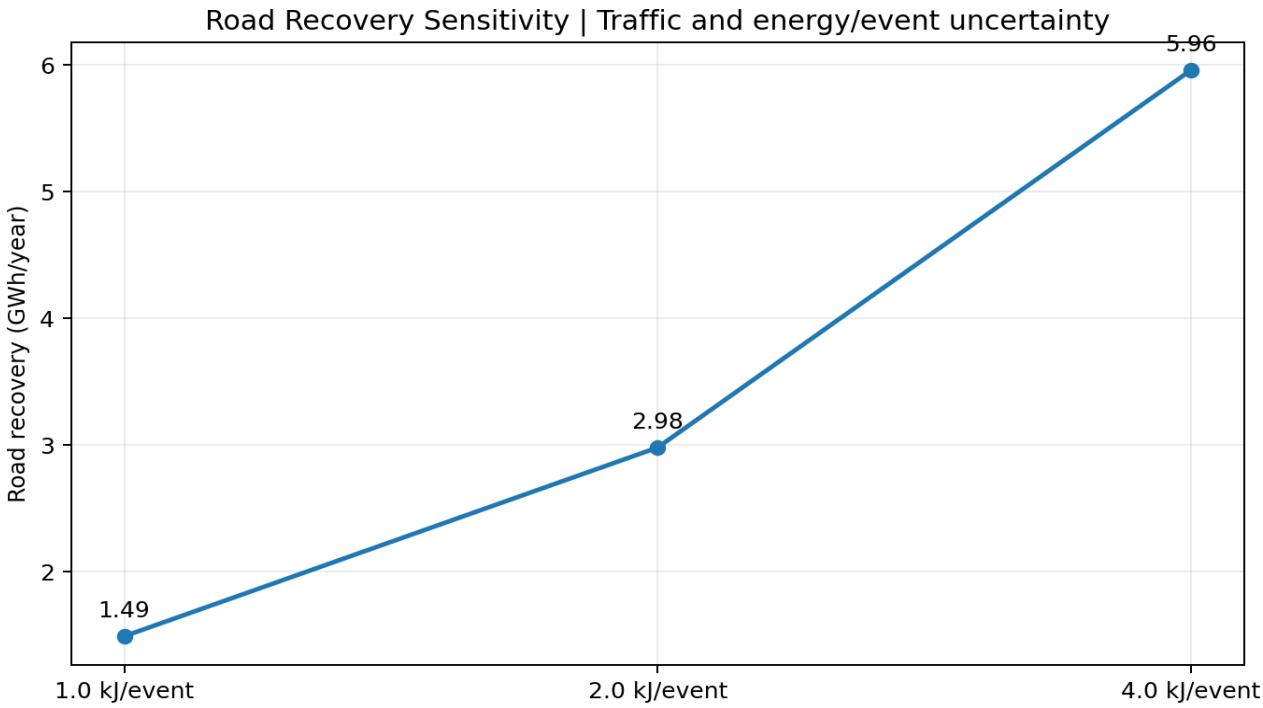
Gross Electricity Value ≠ Profit

The deductions are explicitly illustrative only and are not a final financial model. Advertising/billboard revenue remains separate from electricity revenue.

14

Sensitivity & Scientific Uncertainty

The road module illustrates why field data matters.



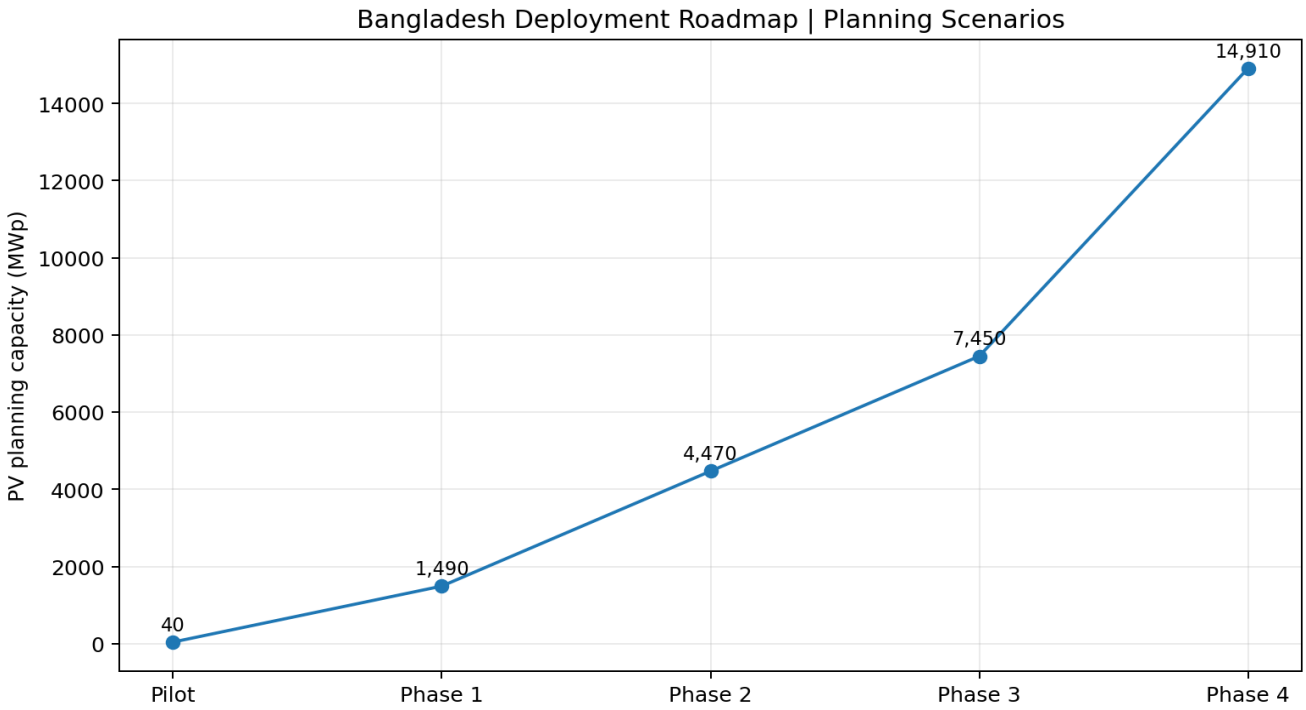
Road sensitivity from the supplied baseline: 1.0 kJ/event → 1.49 GWh/year → Tk 1.55 crore; 2.0 kJ/event → 2.98 GWh/year → Tk 3.10 crore; 4.0 kJ/event → 5.96 GWh/year → Tk 6.20 crore.

Traffic count and energy/event are major uncertainties. Therefore: FIELD VALIDATION BEFORE NATIONAL SCALE-UP.

15

Bangladesh Deployment Roadmap

Scale only after corridor-level engineering, resource and approval checks.



Stage	Length	PV	Solar / road benchmark	Note
Pilot	100 km	40 MWp PV	2.98 GWh road recovery	10 river units
Phase 1	3,727 km	1.49 GWp PV	2,087 GWh solar	Planning scenario
Phase 2	11,181 km	4.47 GWp PV	6,261 GWh solar	Planning scenario
Phase 3	18,635 km	7.45 GWp PV	10,436 GWh solar	Planning scenario
Phase 4	37,270 km	14.91 GWp PV	20,871 GWh solar	Planning scenario

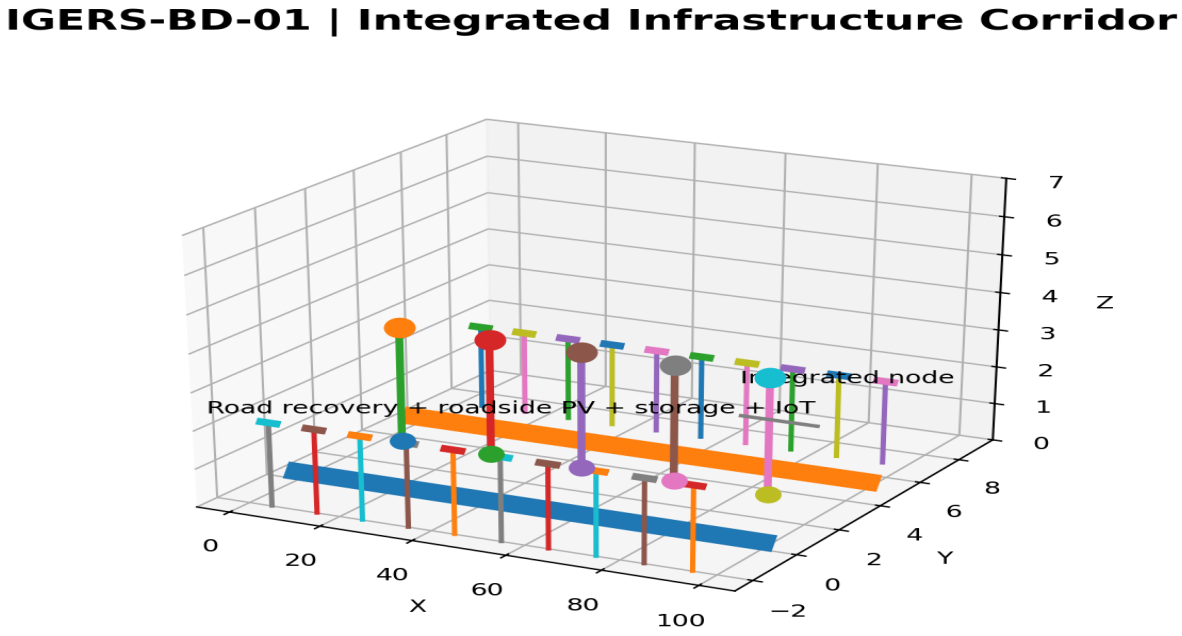
Required before implementation: corridor-by-corridor traffic, land availability, PV setbacks, structural design, shading, hydrology, environmental constraints and approvals.

16

3D Engineering Model Gallery

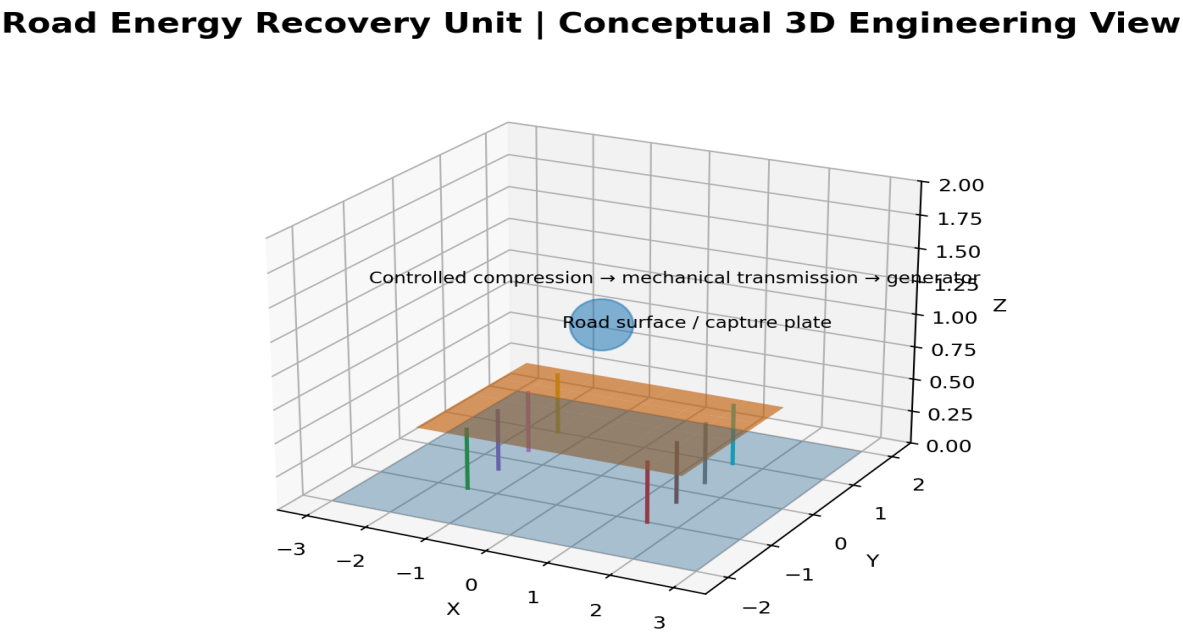
Conceptual visualizations for system communication and thesis presentation.

01 • Complete IGERS road corridor



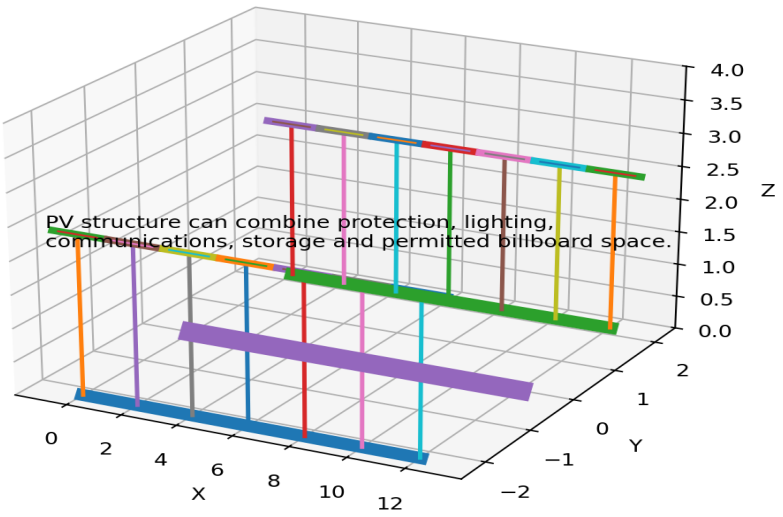
Conceptual engineering visualization — not a construction drawing.

02 • Road energy recovery unit



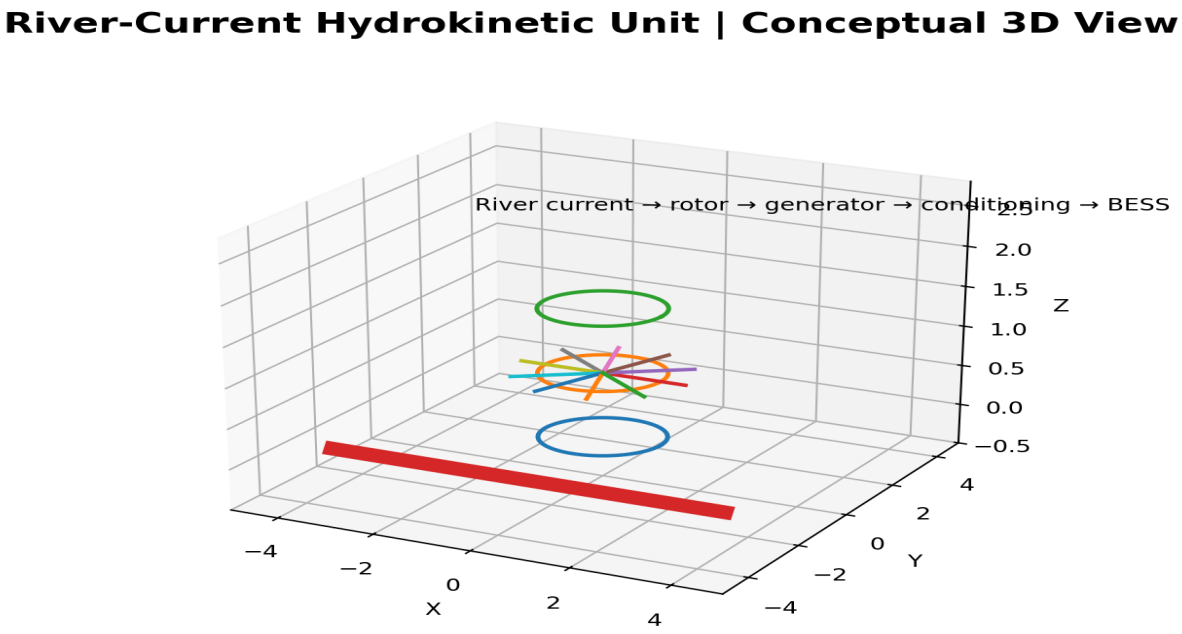
Conceptual engineering visualization — not a construction drawing.

03 • Roadside PV safety barrier



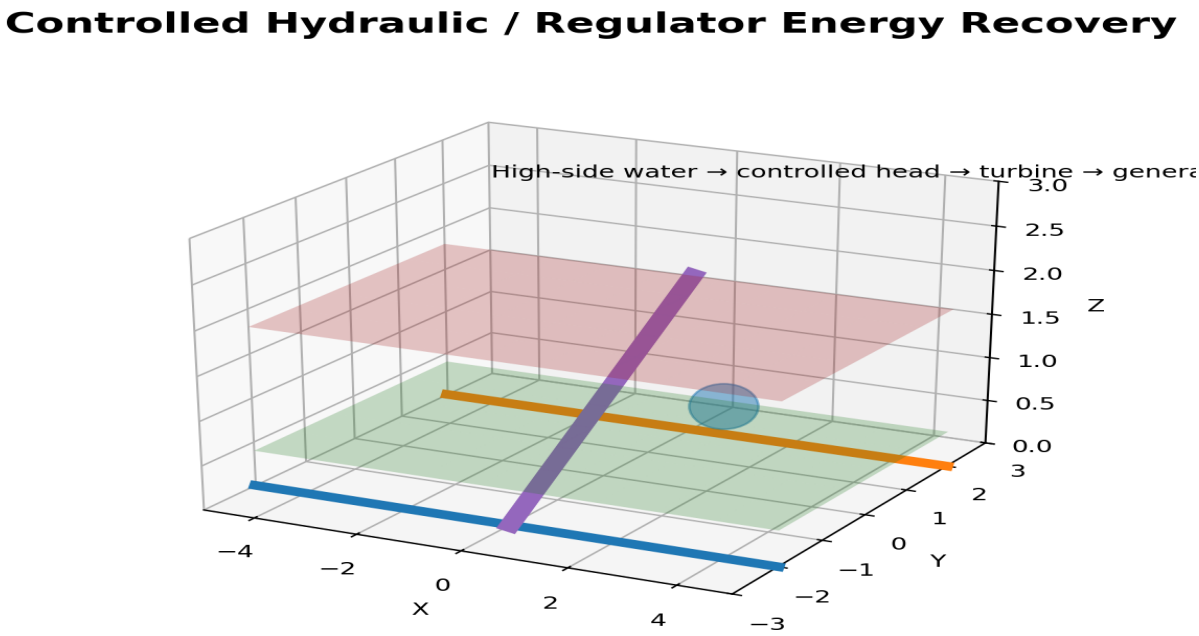
Conceptual engineering visualization — not a construction drawing.

04 • River-current turbine



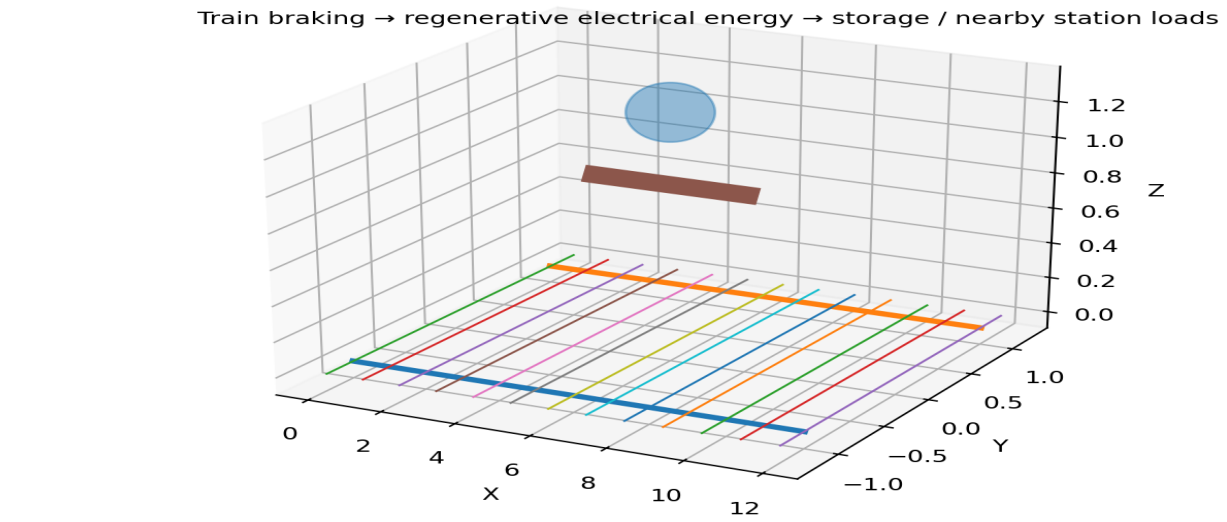
Conceptual engineering visualization — not a construction drawing.

05 • Regulator / low-head turbine



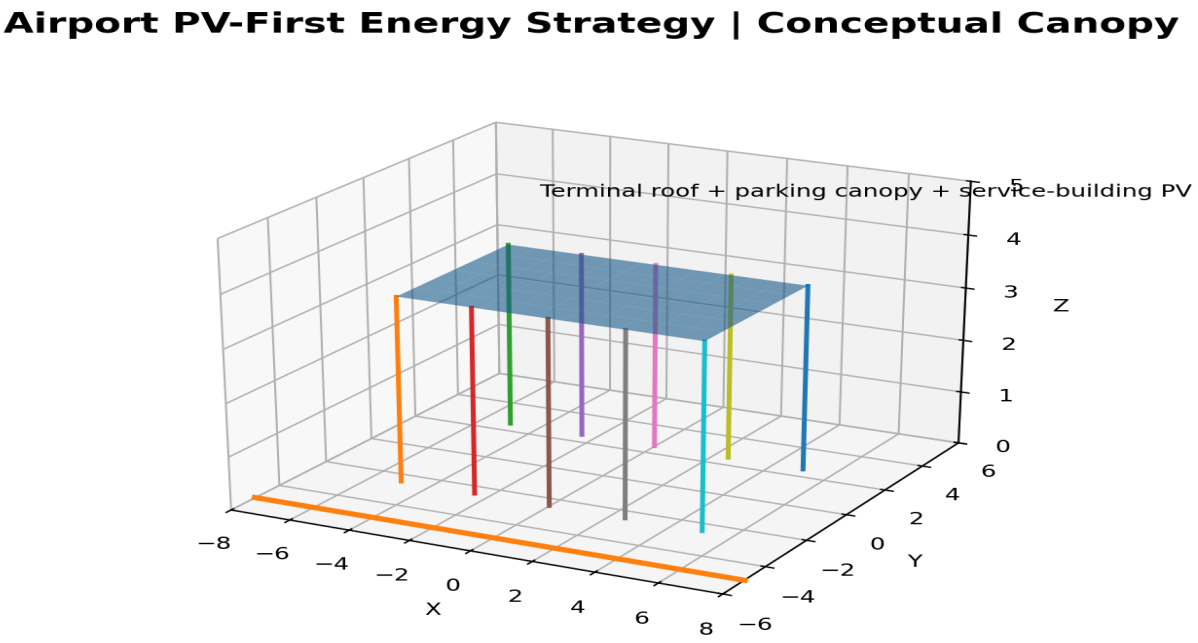
Conceptual engineering visualization — not a construction drawing.

06 • Railway regenerative system



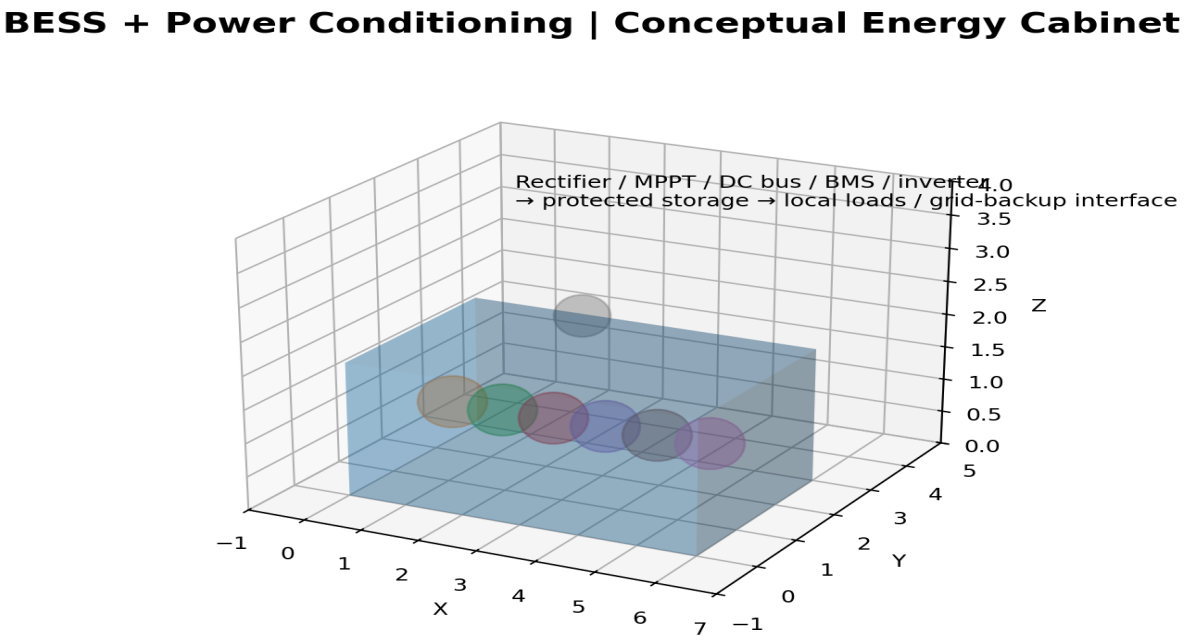
Conceptual engineering visualization — not a construction drawing.

07 • Airport PV canopy



Conceptual engineering visualization — not a construction drawing.

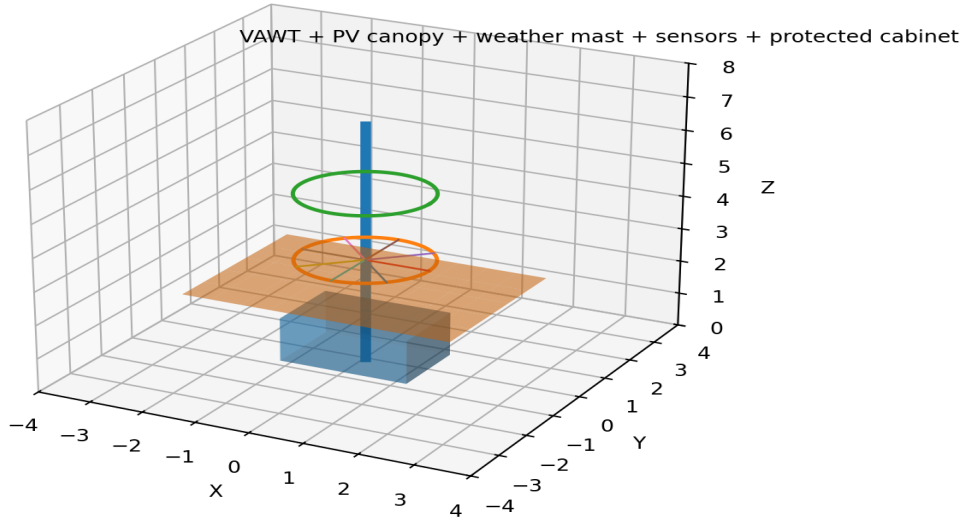
08 • BESS container / energy cabinet



Conceptual engineering visualization — not a construction drawing.

09 • Hybrid solar + VAWT node

Hybrid Solar + VAWT Resilience Node | Conceptual Prototype



Conceptual engineering visualization — not a construction drawing.

17

Field Validation Programme

Convert planning assumptions into measured engineering evidence.

System	Minimum validation dataset
ROAD	AADT; vehicle class; speed distribution; axle load; force-displacement; energy/event; fatigue; safety
SOLAR	GPS; azimuth; tilt; shading; irradiance; soiling; temperature; inverter efficiency; BESS efficiency
WATER	Velocity distribution; flow; head; bathymetry; sediment; debris; seasonal variation; navigation
RAIL	Train mass; speed; stops; regenerative fraction; electrification; receptivity; storage
AIRPORT	Roof/canopy area; glare; obstacles; grid connection; load profile; certification
ECONOMICS	CAPEX; civil works; battery replacement; O&M; insurance; financing; ROW; PPA/wheeling

18

Scientific Defensibility & Limitations

Mature technologies provide the backbone; measured modules provide the research frontier.

Evidence tier	Included technologies
HIGH-CONFIDENCE TECHNOLOGIES	Solar PV; BESS; standard power electronics; conventional local electrical loads
SITE-DEPENDENT TECHNOLOGIES	Road kinetic recovery; river-current turbines; hydraulic recovery; regenerative rail energy; wind turbines
EXPERIMENTAL / MICRO-POWER	Piezo / footstep; track micro-harvesting

Scientific position

The project is strongest when it combines mature technology with measured experimental modules. Conceptual values should remain explicitly labeled until field validation establishes performance, durability and economic viability.

19

Author Profile & CV

Professional profile using the exact supplied project credentials.

YOUR PROFESSIONAL PHOTO

Full Name	Abdullah Al Rafi
Education	B.Sc. in Civil Engineering — Bangladesh Army University of Science and Technology (BAUST)
Current Level	L/T – 3/2
HSC	Bangabandhu College, Rajshahi, Vhadra • 2022 • GPA 4.69 / 5.00
SSC	Green View High School • 2020 • GPA 4.89 / 5.00
B.Sc. CGPA	2.55 / 4.00
Date of Birth	01 March 2004
Address	Gopalganj–Muksudpur
Phone	01898400748
Email	rafimunshi46@gmail.com

Achievements

• Talent Scholarship Recipient — Class 5, through BGB Headquarters, Sarilaka
• Talent Scholarship Recipient — Class 8, through BGB Headquarters, Sarilaka

Technical Skills

• Engineering Design & Analysis
• AutoCAD / 2D Drafting
• Renewable Energy System Concepts
• Energy Recovery & Storage System Design
• Project Planning & Feasibility Analysis
• Microsoft Word
• Microsoft PowerPoint
• Sustainable Infrastructure & Smart Road Concepts

20

Thesis Contribution & Attribution

A system-level contribution built around infrastructure, energy and validation.

Formal attribution

IGERS-BD-01 is attributed to Abdullah Al Rafi [BD].

The project does not falsely claim to have invented PV, batteries, turbines, regenerative braking, piezoelectric harvesting or other established technologies.

Originality is presented as the system-level integration, infrastructure-energy architecture, Bangladesh-focused deployment framework, measurement methodology, storage/control strategy and feasibility framework.

Substantive reproduction, publication or implementation of the IGERS-BD-01 concept should seek permission from the inventor/author.

21

References & Source Integrity

A source-grounded portfolio with no invented external claims.

Primary source basis

This magazine was constructed from the supplied IGERS-BD-01 project specification in the attached file. The numerical assumptions, terminology, architecture, limitations, validation variables and attribution language in this portfolio are drawn from that supplied material.

Source-control rule

Where the source itself identifies a value as illustrative, conceptual, scenario-based or a planning benchmark, that status is retained. No external national measurement is introduced into the project baseline.

End of portfolio

IGERS-BD-01 • Integrated Infrastructure–Energy Recovery & Resilience Architecture • 2026 Professional Thesis & Engineering Concept Edition

22

Energy Accounting Snapshot

Supporting pages for thesis review and professional presentation.

Module	Annual energy	Status
Road recovery	2.98 GWh/year	Controlled fraction; field validation required
Roadside PV	56.00 GWh/year	0.40 MWp/km; 1,400 kWh/kWp-year
River current ×10	1.16 GWh/year	Scenario-based
Rail	0.016 GWh/year	20 trains/day example
Footstep	0.000 GWh/year	Micro-power
Total	60.145 GWh/year	Modeled / planning benchmark

22

Module Interdependency Matrix

Supporting pages for thesis review and professional presentation.

Road recovery	Power conditioning	BESS	Local loads
Roadside PV	MPPT / inverter	BESS	Lighting / communications
Water recovery	Rectifier / generator	BESS	Monitoring / local infrastructure
Rail regeneration	Traction interface	BESS	Station loads
Hybrid node	Rectifier / MPPT	Battery + BMS	LED / telemetry / sensing

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Safety & Structural Design Gates

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- Road: safety and ride comfort before energy yield.
- Bridge: protect the primary structural load path.
- Water: preserve navigation, flood management and hydraulic function.
- Airport: preserve runway friction, braking, drainage, FOD safety and certification.
- BESS: include BMS, protection, thermal management and fire safety.
- Hybrid node: verify foundation, environmental exposure, access and electrical protection.

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Measurement-to-Scale Workflow

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01 Screen site

02 Measure resource/load

03 Prototype

04 Instrument

05 Validate energy/event

06 Assess fatigue/safety

07 Update economics

08 Pilot

09 Review authority requirements

10 Scale corridor-by-corridor

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Engineering Review Checklist

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- Civil/structural integrity
- Electrical protection
- Power conditioning
- Storage sizing
- Environmental constraints
- Maintenance access
- Monitoring/telemetry
- Economic assumptions
- Approvals and authority review
- Documented uncertainty

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Concept Status Legend

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CONCEPTUAL — engineering starting point

ILLUSTRATIVE — example calculation

SCENARIO-BASED — depends on site/resource

PLANNING BENCHMARK — not guaranteed production

MICRO-POWER — sensor/autonomy role

SAFETY-CRITICAL RESEARCH — airport runway harvesting

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100 km Scenario Assumptions

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- 100 km reference corridor
- 210,000 vehicle passages/day
- 1 harvesting point/km
- 0.40 MWp/km roadside PV
- 1,400 kWh/kWp-year
- 10 river-current units in pilot scenario
- Road recovery 2.98 GWh/year
- Roadside solar 56 GWh/year
- Total 60.145 GWh/year
- Tariff benchmark Tk 10.40/kWh

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